Working with learning management systems

(Moodle, Canvas, Google Classrooms, etc.)

What is a learning management system (LMS)?

A learning management system is a software application or web-based technology used

to plan, implement and assess a specific learning process.

It's used for e-learning practices and, in its most common form, consists of two elements: a server that performs the base functionality and a user interface (UI) that is operated by instructors, students and administrators.

Typically, an LMS provides an instructor with a way to create and deliver content, monitor student participation, and assess student performance.

It might also provide students with interactive features, such as threaded discussions, video conferencing and discussion forums.

What are learning management systems used for?

LMSes are beneficial to a range of organizations, including higher education institutions and companies.

They're primarily used for knowledge management: the gathering, organizing, sharing and analysis of an organization's knowledge in terms of resources, documents and people skills.

The role of the LMS varies according to the organization's training strategy and goals.



LMS features

Responsive design: Users can access the LMS from any type of device, whether it's a desktop, laptop, tablet or smartphone. The system automatically displays the version best suited for each user's chosen device and lets users download content for offline work.

User-friendly interface: The UI lets learners navigate the LMS platform and is aligned with the abilities and goals of the user and the organization. An unintuitive UI risks confusing or distracting users and will make the LMS less effective.

Reports and analytics: E-learning assessment tools show instructors and administrators how effective online training initiatives are. Both groups of learners and individuals can be analyzed with these tools and metrics.

LMS features

Catalog and course management: Admins and instructors manage the catalog of course content in the LMS to create more targeted learning experiences.

Content interoperability and integration: Content created and stored in an LMS must be packaged in accordance with interoperable standards, including:

SCORM (Sharable Content Object Reference Model): a widely used standard that ensures online learning content and LMSs can communicate — for example, tracking completion, scores, and time spent. and

xAPI (Experience API or Tin Can API): a newer and more flexible standard that allows tracking of learning experiences beyond the LMS (e.g., mobile learning, simulations, or offline activities).

Support services: Different LMS vendors offer varying levels of support. Many provide online discussion boards where users can connect and help each other. Additional support services, such as a dedicated, toll-free phone number, might be available for an extra cost.

LMS features

Social learning capabilities: Many LMSes include social media tools in their learning platforms to let users interact with their peers, collaborate and share learning experiences.

Gamification: Some LMSes include game mechanics or built-in gamification features that add extra motivation and engagement to courses. This gives students additional incentive to complete courses, in the form of leaderboards, points and badges.

Automation: Learning management systems automate repeated and tedious tasks, such as grouping, adding and deactivating users, and handling group enrollments.

LMS features

Localization: LMSes often include multilingual support, removing language barriers from learning and training content. Some LMSes integrate geolocation features that automatically present the appropriate version of the course when a user accesses it.
For example, an online course can be available in English, Urdu, and Arabic.

Artificial intelligence: LMSes use AI to create personalized learning experiences for users with course formats suited to their needs. AI also helps suggest topics a user might find interesting based on the courses they've already completed.

Certification and compliance support: This feature is essential to systems used for online compliance training and certifications. Instructors and admins assess an individual's skill set and identify any gaps in their performance. This feature also makes it possible to use LMS records during an audit.

Types of LMS deployments

Cloud-based: LMSes are hosted on the cloud and often follow a software as a service (SaaS) business model. Providers maintain the system and handle updates or upgrades. Online users can access the system apps from anywhere at any time using a username and password. [e.g. MoodleCloud, Google Classroom]

Self-hosted: LMSes require the organization to download and install the LMS software. The self-hosted platform provides creative control and customization, but the organization is responsible for maintaining the system and might also have to pay for updates. [e.g. Moodle self-hosted]

Third-party hosted: LMSes are also learning resources hosted by a third-party organization. Courses can be obtained directly from a public cloud location, or from the training company's own data center or private cloud.

Desktop application LMSes are installed on the user's desktop. However, the application might still be accessible on multiple devices.

Mobile application LMSes support a mobile learning environment and are accessible wherever and whenever through mobile devices. This platform deployment type lets users engage with and track their online learning initiatives on the go.

LMS Type	Description	Accessibility	Customization	Examples
Cloud-based	Hosted on the cloud; usually follows a SaaS (Software as a Service) model. Provider handles updates, backups, and maintenance.	Accessible anytime, anywhere via web browser or app with username & password.	Limited; based on provider's options.	MoodleCloud, Google Classroom, Canvas, Blackboard Learn Ultra
Self-hosted	Organization downloads, installs, and runs the LMS on its own server or local system.	Accessible within organization's network or online (if configured).	High — full creative and technical control.	Moodle (self-hosted), Totara Learn, Open edX, Sakai
Third-party hosted	Hosted and maintained by an external (third-party) company or training provider on their own cloud/data center.	Access via vendor's hosted environment (public or private cloud).	Moderate — depends on service agreement.	Docebo, SAP Litmos, Blackboard Managed Hosting
Desktop application	Installed directly on a user's desktop computer; may offer local data storage and offline access.	Limited to installed devices (may support syncing across multiple devices).	Moderate — depends on software.	Forma LMS (desktop version), Easy LMS (desktop)
Mobile application	Designed for smartphones and tablets; supports learning on the go.	Accessible anytime, anywhere via mobile apps.	Limited — focused on mobile-friendly use.	TalentLMS app, Moodle Mobile, Blackboard Mobile Learn

Benefits of an LMS

The ability to monitor users' learning progress and performance.

Increased e-learning accessibility without geographic limitations.

Personalized online courses, training and learning experiences.

The ability to easily and efficiently update e-learning modules and activities.

Consistent and easy distribution of online training and learning content across an organization.

Elimination of repetitive tasks, such as user enrollment and certification.

Centralized learning that lets an organization organize and store all data in one place, making it easier for instructors and admins to update and maintain learning materials.

Advanced encryption features to keep data and content secure.

Moodle LMS

Moodle is a Learning Management System (LMS) used to manage, deliver and measure training and learning online.

Moodle is perhaps the most well-known LMS platform of its kind and is certainly widely used. Launched in 2002, it provides a way for those who manage training to deliver it flexibly and efficiently to different audiences.

Moodle LMS allows you to build and upload e-learning content, deliver it to your learners, assess them on that content, track their progress and recognise their achievements.

Benefits of Moodle LMS

Open-source: It means the code in which it's built is free. The advantages of open-source include: No license fee, No vendor lock, Easily scalable.

Easy to use: Functionality from simple drag-and-drop course builders to progress tracking makes every step of your e-learning strategy simpler and quicker than any offline system and most other LMS options too.

Flexibility: Moodle's flexibility is one of its key strengths. It can be configured, whether using the standard functionality, additions such as plugins or managed solution services, to meet your organization's requirements.

Multi-device compatibility: Moodle LMS can be accessed on any device and is mobile responsive, meaning your users will still get the same user experience regardless of which device they access Moodle on.

Benefits of Moodle LMS

Integration options: Moodle can often be integrated seamlessly with platforms such as Microsoft Teams or Zoom for video conferencing functionality, as well as data integrations with existing HR or CRM systems.

Collaborative nature: Moodle encourages collaboration through forums, Wikis, chat functionality and groups, so your learners have lots of collaborative areas to share thoughts and ideas, as well as work together on projects.

Who uses Moodle LMS?

Educational Organizations

Organizations and Corporations

Used for employee training, compliance courses, and professional development.

Government and NGOs

Governments and NGOs use Moodle for capacity-building programs and e-learning projects.

Private Training Providers

Professional trainers and certification bodies use Moodle for delivering online courses, certifications, and skills training.

Canvas LMS

Canvas is a web-based learning management system, or LMS. It is used by learning institutions, educators, and students to access and manage online course learning materials and communicate about skill development and learning achievement.

Canvas LMS is an open and reliable web-based software that allows institutions to manage digital learning, educators to create and present online learning materials and assess student learning, and students to engage in courses and receive feedback about skill development and learning achievement.

Basic Canvas Features

Instructors may create and share course content using Assignments, Discussions, Modules, Quizzes, and Pages. They may also choose to foster a collaborative learning experience using Collaborations, Conferences, and Groups. Depending on course settings, students can access these areas in Canvas to find learning materials and interact with other course users.

Instructors can provide students with comprehensive feedback on assignment and quiz submissions using SpeedGrader and manage grade reporting in the Canvas Gradebook. They can also facilitate real-time course interactions using Chat and communicate course news and updates with students using Announcements as well as the Calendar and Syllabus.

Canvas Users

Admins

Designers

Instructors

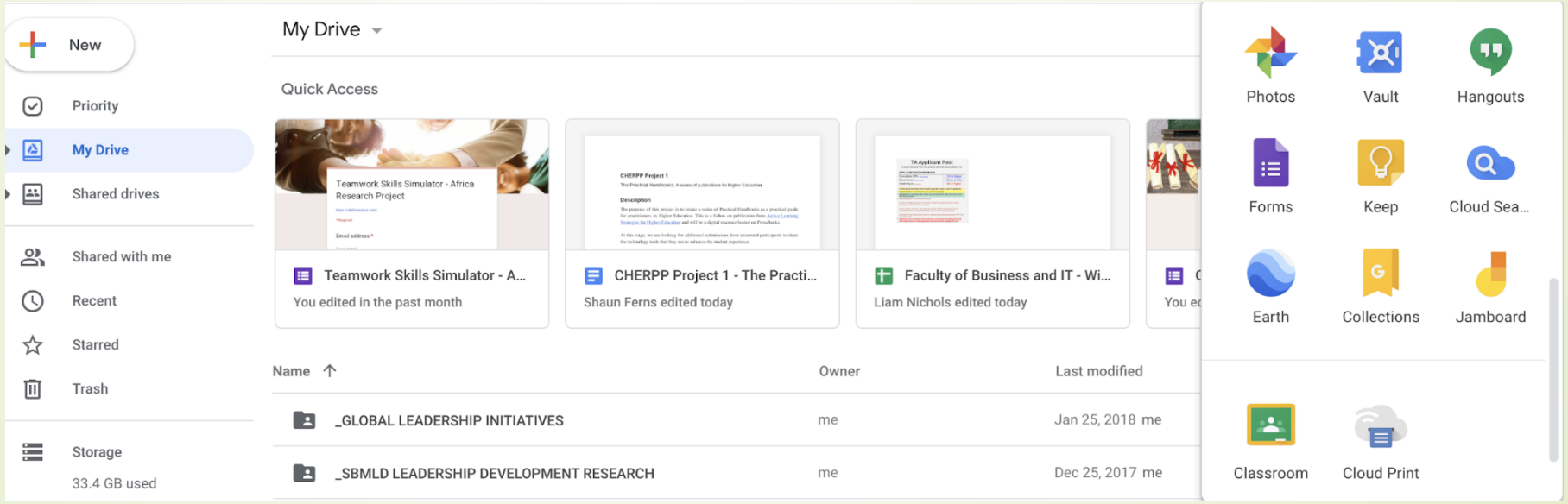
Teacher Assistants

Students

Observers

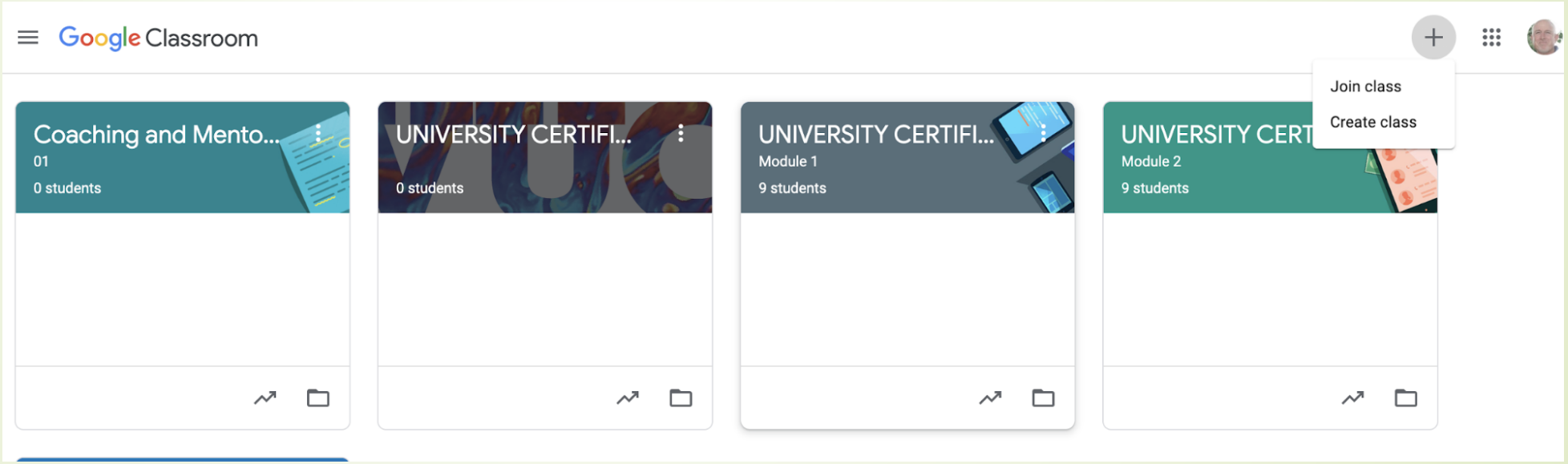
Google Classroom as a Learning Management System

Step 1. **Navigate to Google Classroom:** Navigate to Google Classroom and sign in using your institutional credentials. You can either search for Google Classroom using Google search or sign in to your Google suite using your institutional credentials.



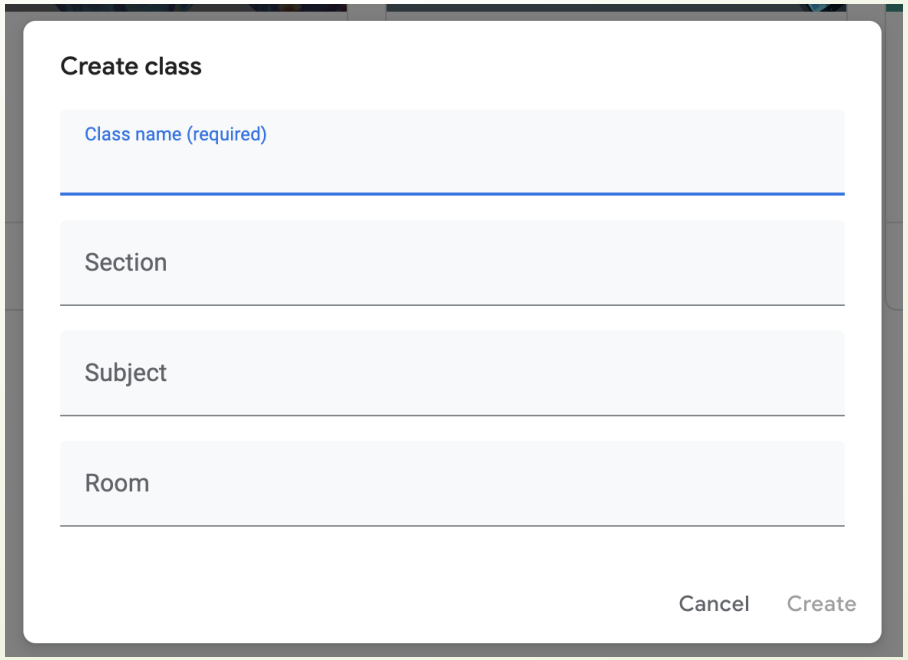
Step 2. Create your course

Click on the + sign in the right-hand corner of the screen to create your course.



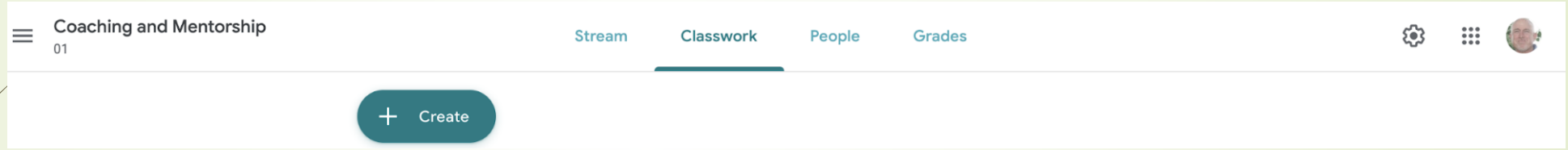
Step 2. Create your course

You will need to name your course and indicate the section of the course as well as the specific subject matter within the course. You can also assign a room number for the course if you have more than one cohort of students enrolled in the course.



Step 3. Create topics within the course.

Click on the + sign to create topics within the course.



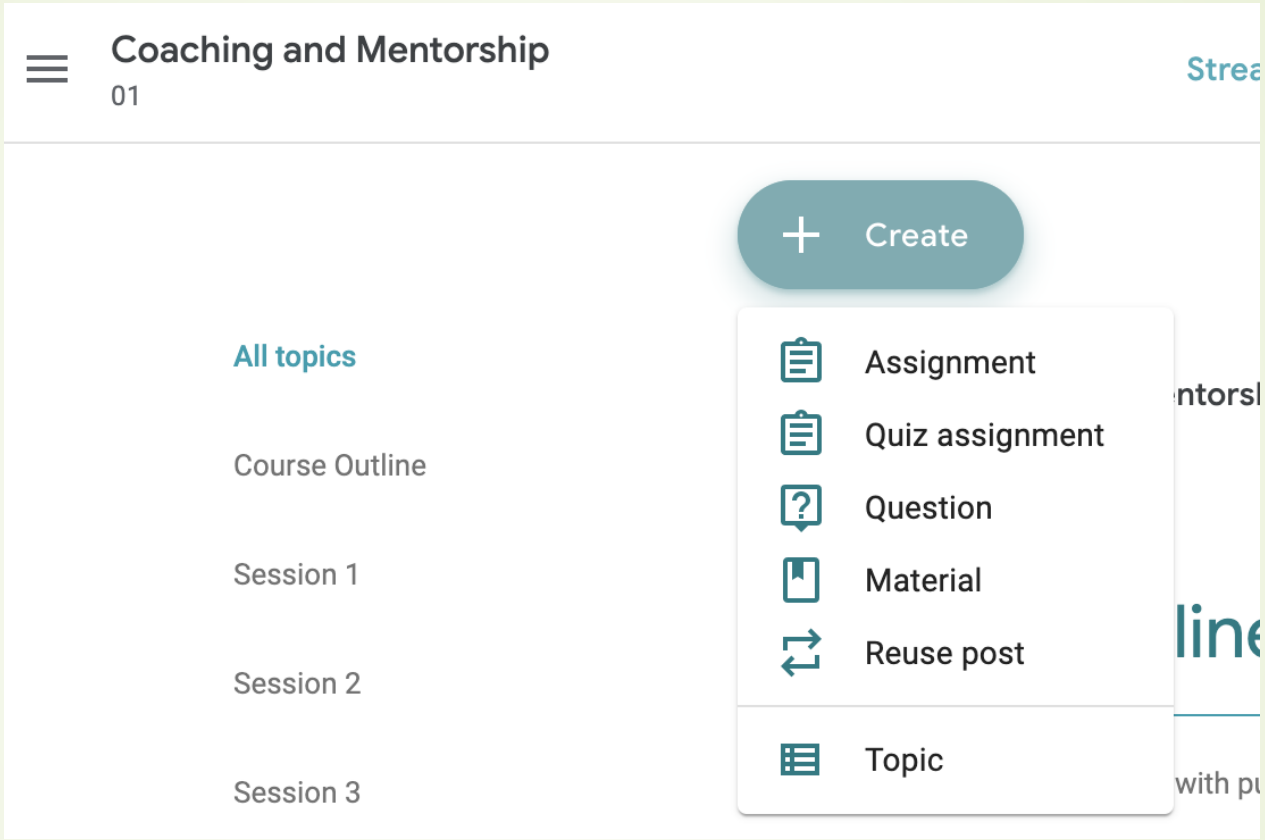
Now that you have created your course you will need to create the topics for your course such as:

Course Outline

Session 1

Session 2

Session 3



You can sequence your modules by easily dragging and dropping each module to the preferred level.

Step 4. Populate your topics

Populate your topics with the material required for student learning. Within your topic you click the + sign and choose to add either:

An Assignment.

A Quiz Assignment.

A Question, or Discussion Question.

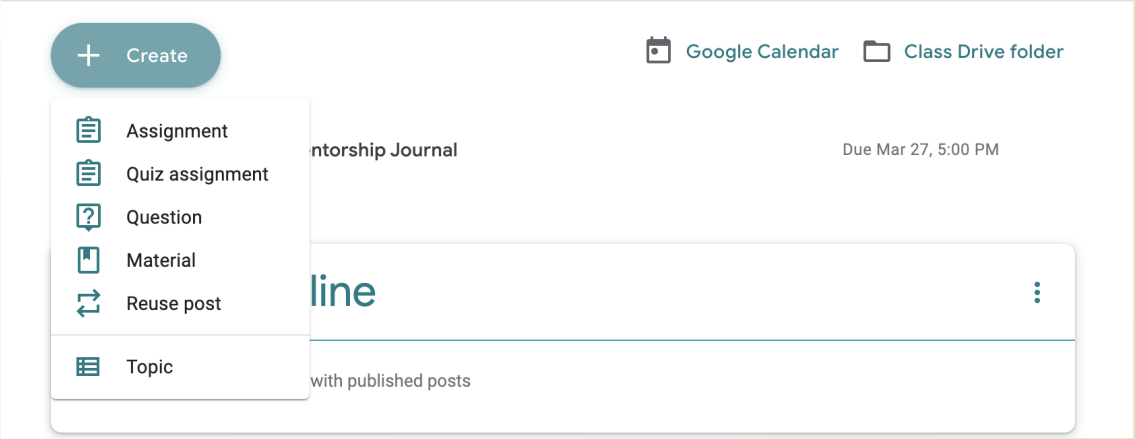
The material comprising either:

A pdf. document.

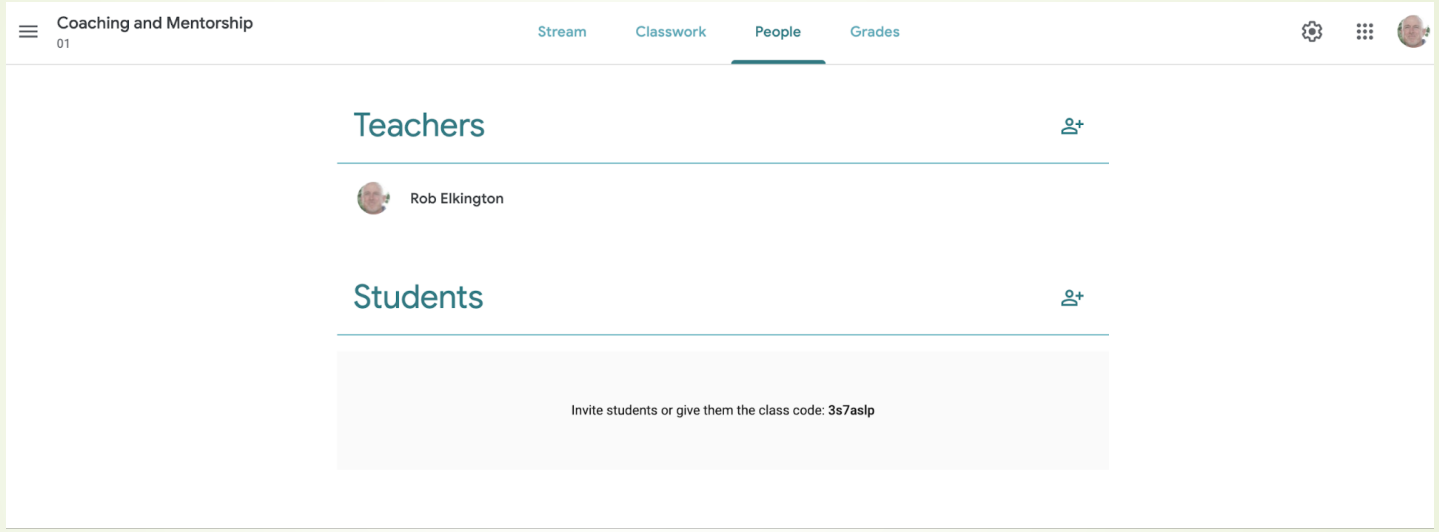
A Word document.

A video clip.

A PowerPoint presentation.



Step 5. Invite students to join the class



Once you have completed the development of your course in Google Classroom you invite students to join either by clicking on the +sign in the “Students” tab or by sending the students an email with the course code included.